KOGI

Independent Worker Operating System

Platform Requirements Document

Functional & Non-Functional Requirements — All Modules

Document Type	Platform Requirements Specification (PRS)
Version	1.0.0 — Initial Release
Status	Active — Under Review
Scope	Complete platform: all 12 modules, all user types
Companion Docs	Part I Architecture · Part II Module Specs · Use Cases & Workflows
Requirement Count	400+ requirements across Functional, Non-Functional, Security, and Compliance categories
Methodology	IEEE 830 / Agile hybrid — each requirement is atomic, testable, and prioritized
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SECTION 1 · OVERVIEW

Introduction & Purpose

1.1 Purpose of This Document

This Platform Requirements Specification (PRS) defines the complete set of functional and non-functional requirements for the Kogi Independent Worker Operating System. It serves as the authoritative reference for product, engineering, design, and QA teams throughout the design, development, and testing lifecycle.

Every requirement in this document is atomic (describing a single capability), testable (verifiable by inspection, demonstration, or test), and prioritized (indicating when it must be delivered). Requirements are grouped by module and cross-referenced where relevant.

1.2 Document Scope

This document covers the full Kogi platform across twelve functional modules, plus cross-cutting concerns including security, performance, compliance, data governance, and external integrations. Requirements for native mobile applications, white-label deployments, and third-party plugin systems are included as future-state items where applicable.

1.3 Requirement Identification

Each requirement is identified with a code in the format:

Format	REQ-[MODULE]-[NUMBER] e.g. REQ-PORT-042
MODULE Codes	KERN · USER · PORT · WBS · MKTG · EXCH · STUD · COMM · WORK · FUND · AO · AI · NFR · SEC · COMP · INT · DATA
Priority Levels	Critical · High · Medium · Low (see legend below)
Status Values	Defined · Approved · Deferred · TBD

CRITICAL — Mandatory, release blocker	**HIGH — Required for v1.0**	**MEDIUM — Target v1.x**	**LOW — Future / Nice-to-have**

SECTION 2 · OVERVIEW

System Overview

2.1 Product Vision

Kogi is a unified operating system for independent workers — a single platform where individuals, teams, cooperatives, and autonomous organizations manage their entire professional and economic lives. Kogi replaces the fragmented stack of project tools, invoicing apps, marketplaces, social networks, and investment dashboards that independent workers currently navigate.

2.2 Platform Architecture Summary

Layer	Components	Primary Language
Kernel	Module registry, event bus, scheduler, provisioner	Zig
API Gateway	REST + gRPC services, authentication, routing	Go
Analytics Engine	Portfolio health, velocity, risk, platform metrics	Scala
Desktop Client	JavaFX native desktop application	Java
Web Client	SPA with Angular Material dark theme	TypeScript/Angular
Mobile Client	Cross-platform React Native application (future)	TypeScript
Build System	Monorepo with multi-language support	Bazel

2.3 Module Inventory

Module	Code	Description	Part
Kernel	KERN	Core orchestrator, event bus, module registry	Part I
User & Workspace	USER	Accounts, workspaces, provisioning, roles	Part I
Portfolio	PORT	Portfolio items, containers, collections, directories	Part I

Module	Code	Description	Part
WBS & Stories	WBS	Work breakdown structure, story types, tasks	Part I
Market System	MKTG	Listings, gigs, services, products, talent discovery	Part II
Exchange System	EXCH	Payments, escrow, equity, barter, distributions	Part II
Idea Studio	STUD	Idea capture, concept dev, prototyping, design	Part II
Community & Social	COMM	Spaces, rooms, chat, feeds, social graph	Part II
Work & Operations	WORK	OKRs, sprints, kanban, automation, capacity	Part II
Crowdfunding & Capital	FUND	Campaigns, pools, stakes, governance	Part I
AO Governance	AO	Autonomous orgs, voting, proposals, charter	Part I
AI Agent	AI	Executor, recommendations, automation, synthesis	Part I+II

SECTION 3 · STAKEHOLDERS

Stakeholders & User Classes

3.1 Primary User Classes

User Class	Description	Primary Modules Used	Scale
Independent Worker	Solo freelancer, consultant, or contractor managing multiple clients and projects	Portfolio, WBS, Market, Exchange, Workspace	Millions
Gig Worker	On-demand task/delivery worker engaged through the platform's gig system	Market (Gig), Exchange, Workspace, Community	Millions
Hobbyist / Enthusiast	Non-commercial user managing personal projects, side ventures, and learning	Portfolio, Studio, WBS, Community	Millions
Small Business Owner	Operator of a small business using Kogi for project and team management	Portfolio, WBS, Work, Market, Exchange	Hundreds of thousands
Team Member	Employee or contractor who is a member of a shared team workspace	Work, WBS, Community, Portfolio	Millions
Cooperative Member	Member of a worker-owned or consumer cooperative governed on the platform	AO, Crowdfunding, Exchange, Market, Work	Hundreds of thousands
Investor / Backer	Individual investing in campaigns, AOs, or cooperative funds	Crowdfunding, Exchange, Portfolio, Market	Hundreds of thousands
AO Contributor	Open source or community contributor to an Autonomous Organization	AO, WBS, Community, Exchange (credits)	Millions
Platform Administrator	Kogi platform staff managing system health, compliance, and disputes	All modules (admin scope)	Hundreds

3.2 Organizational Actors

Team (2–25)	A small group with a shared workspace. Lightweight governance. Shared project and task visibility.
Cooperative (10–500)	A member-owned organization with democratic governance. Shares resources, capital, and decision-making.
Autonomous Organization (AO)	A fully governed entity with charter, capital pools, contributor network, and on-chain audit trail. 10–10,000 members.
Enterprise (future)	Large organization with SSO, custom governance, dedicated infrastructure, and audit/compliance tooling.

SECTION 4 · CONSTRAINTS

Assumptions, Constraints & Dependencies

4.1 Assumptions

- All users have access to a modern web browser (Chrome 110+, Firefox 115+, Safari 16+, Edge 110+) or the native desktop/mobile client.
- Users are responsible for providing their own payment method for Exchange transactions.
- The platform's AI Agent capabilities depend on integration with one or more external LLM providers (e.g., Anthropic Claude, OpenAI GPT).
- Cooperative and AO governance features assume users understand basic voting and democratic process concepts.
- Real-time features (chat, collaborative editing, live presence) require a stable internet connection.
- All financial transactions are subject to the laws and regulations of the user's jurisdiction.

4.2 Constraints

Constraint	Description	Impact
Regulatory	Financial transaction features (escrow, equity, distributions) must comply with applicable securities, money transmission, and payment laws per jurisdiction	Limits automatic feature availability in some markets
Infrastructure	The platform must be deployable on standard cloud infrastructure (AWS, GCP, Azure) without proprietary dependencies	Constrains some real-time and ML feature choices
Accessibility	All web and mobile interfaces must meet WCAG 2.1 AA accessibility standards	Constrains UI design decisions
Data Residency	Enterprise and cooperative customers may require data residency in specific geographic regions	Requires multi-region deployment architecture
Open Source	The kernel and core data models are planned for open-source release; proprietary features must be cleanly separated	Constrains coupling between kernel and commercial modules
Monorepo Build	All code managed in a Bazel monorepo; all modules must conform to Bazel build targets	Constrains technology choices and dependency management

4.3 External Dependencies

Dependency	Type	Module(s)	Criticality
LLM Provider (Anthropic/OpenAI)	AI API	AI Agent (all modules)	Critical
Payment Processor (Stripe)	Financial API	Exchange System	Critical
Identity Provider (Auth0 / custom)	Auth API	User System	Critical
Email Delivery (SendGrid / SES)	Notification API	All modules	High
Push Notifications (Firebase / APNs)	Mobile API	Community, Work, Market	High
Search Engine (Elasticsearch)	Search Infrastructure	Market, Community, Portfolio	High
Object Storage (S3-compatible)	File Storage	Studio, Portfolio, Community	High
Real-time Transport (WebSocket / NATS)	Messaging Infrastructure	Community, Work, Market	High
KYC/AML Provider	Compliance API	Exchange System	High (financial features)
Tax Reporting (Avalara / TaxJar)	Compliance API	Exchange System	Medium
Video Conferencing (WebRTC / Daily.co)	Media API	Community (Video Rooms)	Medium

SECTION 5 · KERNEL

Functional Requirements — Kernel & Platform Core

The Kernel is the central orchestrator of the Kogi platform. All other modules register with and communicate through the Kernel's module registry and event bus.

5.1 Module Registry

ID	Priority	Requirement Description	Status	Notes
REQ-KERN-001	Critical	The system shall maintain a registry of all installed platform modules with their name, version, status, and health metadata.	Defined	
REQ-KERN-002	Critical	The Kernel shall allow modules to register, start, pause, and stop independently without requiring a full platform restart.	Defined	
REQ-KERN-003	Critical	The Kernel shall detect and report module failures, automatically attempting restart up to 3 times before marking the module in error state.	Defined	
REQ-KERN-004	High	The Kernel shall provide a module dependency graph ensuring dependent modules start in correct order.	Defined	
REQ-KERN-005	Medium	The Kernel shall expose a health endpoint listing the status of all registered modules.	Defined	

5.2 Event Bus

ID	Priority	Requirement Description	Status	Notes
REQ-KERN-010	Critical	The platform shall implement a publish-subscribe event bus through which all modules communicate asynchronously.	Defined	
REQ-KERN-011	Critical	The event bus shall guarantee at-least-once delivery for all Critical-category events with configurable retry policy.	Defined	
REQ-KERN-012	Critical	Every event shall carry: event_id, event_type, source_module, payload, timestamp, and correlation_id.	Defined	
REQ-KERN-013	High	The event bus shall support pattern-based subscriptions (e.g., "portfolio.*") in addition to exact-match subscriptions.	Defined	
REQ-KERN-014	High	The system shall maintain an event audit log with configurable retention (default: 90 days) for all events.	Defined	

ID	Priority	Requirement Description	Status	Notes
REQ-KERN-015	High	The event bus shall support event replay for specified time windows to enable module recovery and backfill operations.	Defined	
REQ-KERN-016	Medium	The system shall provide a dead-letter queue for events that fail delivery after maximum retry attempts.	Defined	
REQ-KERN-017	Medium	The event bus shall support ordering guarantees within a single entity's event stream (e.g., all events for portfolio_id:XYZ in order).	Defined	

5.3 Scheduler & Provisioner

ID	Priority	Requirement Description	Status	Notes
REQ-KERN-020	Critical	The Kernel scheduler shall trigger time-based automation rules (recurring tasks, OKR check-ins, contract renewals) with second-level precision.	Defined	
REQ-KERN-021	Critical	The provisioner shall create a complete user environment (Account, Workspace, Portfolio) within 5 seconds of registration.	Defined	
REQ-KERN-022	High	The provisioner shall support configurable workspace templates applied at provisioning time (e.g., freelancer template, startup template).	Defined	
REQ-KERN-023	High	The Kernel shall support multi-tenancy, isolating all data and processing between tenant organizations (for enterprise/cooperative deployments).	Defined	

SECTION 6 · USER

Functional Requirements — User, Account & Workspace

6.1 Account Management

ID	Priority	Requirement Description	Status	Notes
REQ-USER-001	Critical	Users shall be able to register with email + password or via OAuth 2.0 providers (Google, GitHub, Apple).	Defined	
REQ-USER-002	Critical	The system shall verify user email addresses before granting full account access.	Defined	
REQ-USER-003	Critical	Users shall be able to reset their password via an authenticated email link with a 60-minute expiry window.	Defined	
REQ-USER-004	Critical	The system shall support multi-factor authentication (TOTP authenticator app and SMS) as an optional account security measure.	Defined	
REQ-USER-005	High	Users shall be able to update their profile: display name, bio, skills, availability status, location, and social links.	Defined	
REQ-USER-006	High	Users shall be able to upload a profile photo with automatic resizing to platform-standard dimensions.	Defined	
REQ-USER-007	High	The system shall support account deactivation (reversible) and permanent deletion with 30-day grace period per GDPR Article 17.	Defined	
REQ-USER-008	Medium	Users shall be able to export all of their personal data in machine-readable format (JSON) within 30 days of request.	Defined	
REQ-USER-009	Medium	The system shall support username/handle changes with automatic forwarding for 90 days from old handle.	Defined	

6.2 Workspace

ID	Priority	Requirement Description	Status	Notes
REQ-USER-010	Critical	Each user shall have exactly one personal Workspace provisioned automatically upon account creation.	Defined	
REQ-USER-011	Critical	The Workspace shall serve as the unified entry point to all platform modules, with navigation accessible from a persistent sidebar.	Defined	

ID	Priority	Requirement Description	Status	Notes
REQ-USER-012	Critical	Users shall be able to create and join additional Workspaces for teams, cooperatives, and AOs.	Defined	
REQ-USER-013	High	The Workspace shall display an activity feed showing recent events across all active projects and communities.	Defined	
REQ-USER-014	High	Users shall be able to customize Workspace layout: module visibility, sidebar ordering, notification preferences, and theme.	Defined	
REQ-USER-015	High	The system shall support role-based access control within shared Workspaces: Owner, Admin, Member, Guest.	Defined	
REQ-USER-016	Medium	Workspace Owners shall be able to configure custom branding (logo, name, color scheme) for their organization's Workspace.	Defined	
REQ-USER-017	Medium	The system shall support Workspace templates providing pre-configured modules, stories, and structure for common use cases.	Defined	

6.3 Roles & Permissions

ID	Priority	Requirement Description	Status	Notes
REQ-USER-020	Critical	The system shall enforce role-based access at the Workspace, Project, and Resource level.	Defined	
REQ-USER-021	Critical	Owners shall have full administrative control. Admins shall be able to manage members and settings. Members shall have standard access. Guests shall have read-only access to explicitly shared items.	Defined	
REQ-USER-022	High	AO and Cooperative Workspaces shall support custom role definitions with configurable permission sets.	Defined	
REQ-USER-023	High	The system shall maintain an audit log of all permission and role changes with actor, timestamp, and previous value.	Defined	
REQ-USER-024	Medium	The system shall support time-limited access grants for external collaborators.	Defined	

SECTION 7 · PORTFOLIO

Functional Requirements — Portfolio System

7.1 Portfolio Structure

ID	Priority	Requirement Description	Status	Notes
REQ-PORT-001	Critical	Each User and AO shall have at least one root Portfolio provisioned automatically.	Defined	
REQ-PORT-002	Critical	A Portfolio shall support the following item types: Program, Project, Sub-Portfolio, and Resource.	Defined	
REQ-PORT-003	Critical	Portfolio items shall support metadata: name, description, status, tags, and custom key-value fields.	Defined	
REQ-PORT-004	Critical	Portfolios shall support hierarchical nesting: Programs contain Projects; Projects contain Processes; any item may contain Resources.	Defined	
REQ-PORT-005	High	The system shall support Portfolio Collections (unordered groupings) and Directories (ordered, schedulable lists).	Defined	
REQ-PORT-006	High	The system shall support Container types attached to any Portfolio item: Binder, Book, Notepad, Folder, and Briefcase.	Defined	
REQ-PORT-007	High	Users shall be able to set item visibility: Private, Team, Organization, or Public.	Defined	
REQ-PORT-008	High	The Portfolio shall support bulk operations: archive, tag, move, and export across multiple items simultaneously.	Defined	
REQ-PORT-009	Medium	The Portfolio shall provide a configurable Dashboard view with user-selected metrics: item count, health score, active projects, and capital summary.	Defined	
REQ-PORT-010	Medium	The system shall support Portfolio sharing: a read-only public view of selected items for client or investor presentation.	Defined	

7.2 Resource Management

ID	Priority	Requirement Description	Status	Notes
REQ-PORT-020	Critical	The system shall support the following Resource types: Artifact, Asset, Capital, Land, Estate, Labor, Investment, and Account.	Defined	
REQ-PORT-021	Critical	Capital resources shall support a balance, currency, and transaction history linked to the Exchange System.	Defined	

ID	Priority	Requirement Description	Status	Notes
REQ-PORT-022	High	Labor resources shall include a rate, availability schedule, and skill tags for matching against WBS story requirements.	Defined	
REQ-PORT-023	High	Asset resources shall support depreciation tracking with configurable schedules (straight-line, declining-balance).	Defined	
REQ-PORT-024	High	Investment resources shall display current value, cost basis, unrealized gain/loss, and linked transaction history.	Defined	
REQ-PORT-025	Medium	The system shall generate an AI-assisted Portfolio Health Score (0–100) based on item status distribution, resource utilization, and goal progress.	Defined	
REQ-PORT-026	Medium	The Portfolio shall support domain-based organization: users may assign items to domains (e.g., Business, Personal, Investments) with separate dashboard views.	Defined	

SECTION 8 · WBS & STORIES

Functional Requirements — WBS & Story System

8.1 Work Breakdown Structure

ID	Priority	Requirement Description	Status	Notes
REQ-WBS-001	Critical	The system shall support a six-level WBS hierarchy: Work Package → Theme → Initiative → Epic → Story → Task.	Defined	
REQ-WBS-002	Critical	Any Portfolio Project or Program shall be able to have an associated WBS.	Defined	
REQ-WBS-003	Critical	Stories shall be the primary unit of executable work and shall support assignment to one or more Executors.	Defined	
REQ-WBS-004	Critical	Stories shall support status transitions: Backlog → Ready → In Progress → In Review → Blocked → Done → Cancelled.	Defined	
REQ-WBS-005	Critical	The system shall enforce that story status transitions are valid (e.g., cannot skip from Backlog to Done without intermediate states unless overridden by owner).	Defined	
REQ-WBS-006	High	The WBS shall support drag-and-drop reordering within any hierarchy level.	Defined	
REQ-WBS-007	High	The system shall display WBS as: List view, Kanban board, Agile board, Gantt chart, and Tree view.	Defined	
REQ-WBS-008	High	The system shall calculate and display story point velocity per sprint and rolling 6-sprint average.	Defined	
REQ-WBS-009	High	The system shall surface dependency relationships between stories and flag circular or broken dependencies.	Defined	
REQ-WBS-010	Medium	The WBS shall support bulk import of stories from CSV with field mapping.	Defined	

8.2 Story Types

ID	Priority	Requirement Description	Status	Notes
REQ-WBS-020	Critical	The system shall support at minimum the following 30+ story types: Feature, Bug, Capability, Enabler, Blocker, Test, Requirement, Defect, Issue, Review, Audit, Report, Performance, Strategy, Tactic, Operation, Business Case, Research, Prototype, Spike, Idea, Concept, Design,	Defined	

ID	Priority	Requirement Description	Status	Notes
		Documentation, Milestone, Objective, Outcome, Risk, Mission, Vision, Goal.		
REQ-WBS-021	Critical	Each story type shall have appropriate default fields, acceptance criteria templates, and completion requirements.	Defined	
REQ-WBS-022	High	Story types shall be filterable, and custom story types shall be creatable at the Workspace level.	Defined	
REQ-WBS-023	High	The system shall enforce that Blocker stories have a linked blocking story and a resolution plan before being moved to Done.	Defined	
REQ-WBS-024	High	Risk stories shall require a probability, impact score, and mitigation plan.	Defined	
REQ-WBS-025	Medium	The system shall auto-generate story summaries via AI Agent for stories with >500 characters of content.	Defined	

8.3 Agile & Sprint Management

ID	Priority	Requirement Description	Status	Notes
REQ-WBS-030	Critical	The system shall support sprint creation with configurable duration (1–4 weeks), goal, and story commitments.	Defined	
REQ-WBS-031	Critical	Sprint planning shall display team capacity vs. committed story points with overload warnings.	Defined	
REQ-WBS-032	High	The system shall generate a sprint burndown chart updated in real time as stories are completed.	Defined	
REQ-WBS-033	High	Sprint close shall automatically generate a velocity calculation, goal achievement status, and retrospective prompt.	Defined	
REQ-WBS-034	High	The AI Agent shall provide an optional sprint planning recommendation based on backlog priority and historical velocity.	Defined	
REQ-WBS-035	Medium	The system shall support parallel sprints across multiple WBS instances within the same Workspace.	Defined	

SECTION 9 · MARKET

Functional Requirements — Market System

9.1 Listings

ID	Priority	Requirement Description	Status	Notes
REQ-MKTG-001	Critical	The system shall allow any authenticated user to create a Market Listing with: type, title, description, pricing, category, tags, and media.	Defined	
REQ-MKTG-002	Critical	Listing types shall include: Service, Gig, Product (digital), Product (physical), Talent, Tool/API, Resource Rental, and Opportunity.	Defined	
REQ-MKTG-003	Critical	Listings shall be searchable via full-text search across title, description, tags, and provider name.	Defined	
REQ-MKTG-004	Critical	The search system shall support faceted filtering: category, price range, location, availability, rating, and listing type.	Defined	
REQ-MKTG-005	High	The system shall display listing search results with relevance ranking combining text match, reputation score, and recency.	Defined	
REQ-MKTG-006	High	Listings shall support rich media: up to 10 images, 1 video, and 5 document attachments.	Defined	
REQ-MKTG-007	High	Providers shall be able to pause, reactivate, and archive their listings without deleting review history.	Defined	
REQ-MKTG-008	Medium	The AI Agent shall generate an optimized listing title and description from the provider's Portfolio and past work history.	Defined	
REQ-MKTG-009	Medium	The system shall surface "Similar Listings" on each listing page using collaborative filtering.	Defined	

9.2 Orders & Transactions

ID	Priority	Requirement Description	Status	Notes
REQ-MKTG-010	Critical	Buyers shall be able to place an Order on any active listing, triggering the contract and escrow flow.	Defined	
REQ-MKTG-011	Critical	Every Order shall automatically create a corresponding Portfolio item in the buyer's workspace.	Defined	
REQ-MKTG-012	Critical	For service and gig orders, a WBS story shall be auto-created in the buyer's active project WBS.	Defined	

ID	Priority	Requirement Description	Status	Notes
REQ-MKTG-013	Critical	Order status transitions shall trigger notifications to both buyer and seller.	Defined	
REQ-MKTG-014	High	The system shall support milestone-based orders where payment releases in tranches upon milestone completion.	Defined	
REQ-MKTG-015	High	Buyers shall be able to submit a custom brief when placing an order, and sellers shall be able to accept or decline with a counter-proposal.	Defined	
REQ-MKTG-016	High	The system shall support recurring orders (retainer contracts) with automatic renewal and invoicing.	Defined	

9.3 Trust & Reputation

ID	Priority	Requirement Description	Status	Notes
REQ-MKTG-020	Critical	Upon order completion, both buyer and seller shall be prompted to leave a rating (1–5 stars) and written review.	Defined	
REQ-MKTG-021	Critical	Provider rating scores shall be displayed on all listing and profile pages and shall be calculated from a minimum of 3 reviews to be publicly shown.	Defined	
REQ-MKTG-022	High	The system shall support provider credential verification: government ID, professional license, skill certification, and portfolio verification.	Defined	
REQ-MKTG-023	High	Verified credentials shall display a visible badge on provider profiles and listings.	Defined	
REQ-MKTG-024	High	Sellers shall be able to respond to reviews publicly.	Defined	
REQ-MKTG-025	Medium	The AI Agent shall flag suspicious review patterns (e.g., bulk same-day 5-star reviews from new accounts) for moderation review.	Defined	

9.4 Discovery & Recommendations

ID	Priority	Requirement Description	Status	Notes
REQ-MKTG-030	High	The system shall display a personalized "Recommended for You" feed on the Market home page based on Portfolio context.	Defined	
REQ-MKTG-031	High	The AI Agent shall proactively surface Market listings when a WBS story is blocked or has an unmet resource requirement.	Defined	

ID	Priority	Requirement Description	Status	Notes
REQ-MKTG-032	High	The system shall maintain saved searches and alert users when new listings match their saved criteria.	Defined	
REQ-MKTG-033	Medium	The system shall support "Request for Proposal" listings where buyers post requirements and sellers submit proposals.	Defined	
REQ-MKTG-034	Medium	The Market shall support a "Trending" section showing the most-viewed and most-hired listings in each category over rolling 7-day and 30-day windows.	Defined	

SECTION 10 · EXCHANGE

Functional Requirements — Exchange System

10.1 Wallet & Payments

ID	Priority	Requirement Description	Status	Notes
REQ-EXCH-001	Critical	Each user account shall have an associated Exchange Wallet with separate balances for: fiat (multi-currency), service credits, and equity holdings.	Defined	
REQ-EXCH-002	Critical	Users shall be able to deposit funds via credit card, debit card, and ACH/bank transfer.	Defined	
REQ-EXCH-003	Critical	Users shall be able to withdraw available funds to a verified bank account within 2 business days.	Defined	
REQ-EXCH-004	Critical	All fiat transaction processing shall be handled by a PCI-DSS Level 1 compliant payment processor.	Defined	
REQ-EXCH-005	High	The system shall support multi-currency wallets with real-time exchange rate display.	Defined	
REQ-EXCH-006	High	The system shall generate downloadable transaction statements (PDF and CSV) for any date range.	Defined	

10.2 Escrow

ID	Priority	Requirement Description	Status	Notes
REQ-EXCH-010	Critical	All Market orders above $10 shall use escrow: buyer funds are held until delivery confirmation.	Defined	
REQ-EXCH-011	Critical	Escrow funds shall be held in segregated accounts separate from operational funds.	Defined	
REQ-EXCH-012	Critical	The buyer shall have a configurable review window (24 hours to 7 days) to inspect delivery before escrow auto-releases.	Defined	
REQ-EXCH-013	Critical	Either party shall be able to open a dispute during the review window, which halts auto-release and initiates the Dispute Resolution process.	Defined	
REQ-EXCH-014	High	The system shall support multi-milestone escrow: funds split across milestones, each released independently upon milestone approval.	Defined	

ID	Priority	Requirement Description	Status	Notes
REQ-EXCH-015	High	The system shall send email and in-app notifications at: escrow funded, delivery submitted, review window open, 24h before auto-release.	Defined	

10.3 Value Types Beyond Fiat

ID	Priority	Requirement Description	Status	Notes
REQ-EXCH-020	High	The system shall support Service Credits earned through: community contributions, referrals, content creation, and AO work.	Defined	
REQ-EXCH-021	High	Service Credits shall be spendable in the Market for eligible listings and shall display a platform-defined fiat equivalent value.	Defined	
REQ-EXCH-022	High	The system shall support Barter Agreements: two parties agree to exchange services of specified scope; each escrows their commitment; mutual completion triggers release.	Defined	
REQ-EXCH-023	High	The system shall record and track equity stakes in AOs, displaying: stake percentage, cost basis, current implied value, and distribution history.	Defined	
REQ-EXCH-024	Medium	The system shall support revenue-share agreements where a percentage of future income from a specified source is automatically routed to a third party.	Defined	
REQ-EXCH-025	Medium	The system shall support IP licensing transactions where rights to use an asset are granted under specified terms with automated royalty tracking.	Defined	

10.4 Distributions

ID	Priority	Requirement Description	Status	Notes
REQ-EXCH-030	Critical	The system shall support automated profit and revenue distributions from AO Capital Pools to member wallets.	Defined	
REQ-EXCH-031	Critical	Distribution calculations shall be based on configurable allocation rules: proportional to equity stake, proportional to contribution, equal share, or hybrid.	Defined	
REQ-EXCH-032	High	AO Governance approval shall be required before any distribution above a configurable threshold (default: $1,000 total).	Defined	
REQ-EXCH-033	High	The system shall generate a distribution ledger entry for each recipient showing calculation method, gross, fees, and net.	Defined	

ID	Priority	Requirement Description	Status	Notes
REQ-EXCH-034	Medium	The system shall generate IRS Form 1099 data exports for US-based recipients receiving $600 or more in a calendar year.	Defined	

SECTION 11 · IDEA STUDIO

Functional Requirements — Idea Studio

11.1 Idea Capture

ID	Priority	Requirement Description	Status	Notes
REQ-STUD-001	Critical	Users shall be able to capture an Idea via text input in under 10 seconds with zero required fields.	Defined	
REQ-STUD-002	Critical	The Studio shall support the following capture modalities: text, voice recording, image/sketch upload, URL import, and email forwarding.	Defined	
REQ-STUD-003	Critical	Each captured Idea shall create an Idea Story in the user's Studio workspace with an auto-generated title if none is provided.	Defined	
REQ-STUD-004	High	Voice recordings shall be automatically transcribed to text within 60 seconds of upload.	Defined	
REQ-STUD-005	High	URL imports shall automatically fetch page title, description, and key content for AI summarization.	Defined	
REQ-STUD-006	High	Uploaded sketch images shall be processed by the AI Agent to generate a textual description and extract visible text.	Defined	
REQ-STUD-007	Medium	The system shall provide a browser extension for one-click idea capture from any webpage.	Defined	
REQ-STUD-008	Medium	The system shall provide an email-to-idea inbox address per workspace (e.g., ideas@[workspace].kogi.io).	Defined	

11.2 Concept & Pipeline Progression

ID	Priority	Requirement Description	Status	Notes
REQ-STUD-010	Critical	The Studio shall implement a five-stage pipeline: Spark → Concept → Research → Prototype → Design.	Defined	
REQ-STUD-011	Critical	Users shall be able to manually advance an Idea to the next pipeline stage at any time.	Defined	
REQ-STUD-012	High	When an Idea is advanced to Concept stage, the AI Agent shall automatically perform a synthesis pass producing: problem reframing, competitor scan, audience identification, viability score (5 dimensions), and 10 key questions.	Defined	

ID	Priority	Requirement Description	Status	Notes
REQ-STUD-013	High	The Studio shall provide guided framework templates at the Concept stage: Lean Business Model Canvas, Value Proposition Canvas, SWOT, Assumptions Register, TAM/SAM/SOM.	Defined	
REQ-STUD-014	High	Users shall be able to park an Idea (removing it from active view) without deleting it.	Defined	
REQ-STUD-015	Medium	The AI Agent shall proactively suggest advancing an Idea when the current stage's minimum criteria are met.	Defined	

11.3 Prototype & Design

ID	Priority	Requirement Description	Status	Notes
REQ-STUD-020	High	The Studio shall support Prototype story types with structured fields: materials list, build steps, tools required, and test criteria.	Defined	
REQ-STUD-021	High	The AI Agent shall generate a Design Brief document from a completed Prototype story, suitable for sharing with manufacturers or designers.	Defined	
REQ-STUD-022	High	The system shall provide an IP Timestamp feature: creating a cryptographically signed, timestamped record of an Idea's description and authorship.	Defined	
REQ-STUD-023	High	When a Design story is marked complete, the AI Agent shall perform an automated design critique covering: usability, market fit, differentiation, feasibility, and accessibility.	Defined	
REQ-STUD-024	Medium	Users shall be able to invite collaborators to a Studio Idea, with configurable access (view, comment, edit).	Defined	
REQ-STUD-025	Medium	Any Idea at Prototype stage or beyond shall be promotable to a full Portfolio Project with a single action, auto-creating a linked WBS.	Defined	

SECTION 12 · COMMUNITY

Functional Requirements — Community & Social

12.1 Spaces

ID	Priority	Requirement Description	Status	Notes
REQ-COMM-001	Critical	The system shall support Spaces as community hubs with: name, description, type, membership rules, and content policy.	Defined	
REQ-COMM-002	Critical	Space types shall include: Public, Private, AO-governed, Project-scoped, Industry-curated, and Learning.	Defined	
REQ-COMM-003	Critical	Any authenticated user shall be able to create a Public or Private Space.	Defined	
REQ-COMM-004	High	Spaces shall support configurable membership: open join, invite-only, application with approval, or AO membership-gated.	Defined	
REQ-COMM-005	High	Space Admins shall be able to manage members: invite, approve applications, assign roles, and remove members.	Defined	
REQ-COMM-006	High	Each Space shall have a configurable set of Rooms, including at minimum a General room.	Defined	
REQ-COMM-007	Medium	The platform shall provide a Space discovery page with search, category browsing, and curated featured Spaces.	Defined	

12.2 Rooms & Chat

ID	Priority	Requirement Description	Status	Notes
REQ-COMM-010	Critical	The system shall support the following Room types: General (async thread), Live (real-time chat), Voice, Video, Showcase, Event, Announcement, and Q&A.	Defined	
REQ-COMM-011	Critical	Real-time message delivery in Live rooms shall achieve < 500ms end-to-end latency under normal load.	Defined	
REQ-COMM-012	Critical	The system shall support threaded message replies up to 5 levels deep.	Defined	
REQ-COMM-013	Critical	Messages shall support: text (Markdown), emoji reactions, file attachments, @mentions, and links to Portfolio items, Market listings, and Studio ideas.	Defined	

ID	Priority	Requirement Description	Status	Notes
REQ-COMM-014	High	The system shall support Direct Messages between any two connected members, and Group DMs for up to 25 participants.	Defined	
REQ-COMM-015	High	Voice Rooms shall support up to 100 concurrent speakers with WebRTC-based audio.	Defined	
REQ-COMM-016	High	Video Rooms shall support up to 25 concurrent video participants with screen sharing capability.	Defined	
REQ-COMM-017	High	Members shall be able to invite the AI Agent (@kogi) into any room; the agent shall respond in-thread within 5 seconds.	Defined	
REQ-COMM-018	Medium	The system shall maintain message history with full-text search within each Room.	Defined	
REQ-COMM-019	Medium	Rooms shall support scheduled events with calendar reminders sent 24 hours and 15 minutes before start.	Defined	

12.3 Social Graph & Feeds

ID	Priority	Requirement Description	Status	Notes
REQ-COMM-020	Critical	The system shall support the following relationship types: Follow (directed), Connection (mutual), Collaborator (project-scoped), Client/Provider (order-scoped), AO Member, Mentor/Mentee, and Cooperative Member.	Defined	
REQ-COMM-021	Critical	Each user shall have a personalized Activity Feed displaying content from followed members, joined Spaces, and AI-curated discovery content.	Defined	
REQ-COMM-022	High	Users shall be able to post public updates, portfolio highlights, and project achievements to their profile and followers' feeds.	Defined	
REQ-COMM-023	High	The system shall provide a member search with filters: skills, location, availability, industry, and cooperative membership.	Defined	
REQ-COMM-024	High	User profiles shall display: bio, skills, availability, portfolio highlights, review scores (if Market seller), Space memberships, and recent activity.	Defined	
REQ-COMM-025	Medium	The AI Agent shall suggest relevant connections to users based on Portfolio content, skills, and mutual spaces.	Defined	
REQ-COMM-026	Medium	The system shall support a Daily AI Digest: a once-daily email or in-app summary of the most relevant Community and platform activity for each user.	Defined	

SECTION 13 · WORK & OPS

Functional Requirements — Work, Strategy & Operations

13.1 OKRs & Strategy

ID	Priority	Requirement Description	Status	Notes
REQ-WORK-001	Critical	The system shall support Objectives and Key Results (OKRs) at four organizational levels: Personal, Team, Cooperative, and AO.	Defined	
REQ-WORK-002	Critical	Each Objective shall support 2–7 Key Results, each with: description, current value, target value, unit, and owner.	Defined	
REQ-WORK-003	Critical	OKRs shall be linkable to Stories, Projects, and Portfolio Programs to trace execution to strategy.	Defined	
REQ-WORK-004	High	The system shall display an OKR cascade view showing how individual OKRs contribute to team and organizational OKRs.	Defined	
REQ-WORK-005	High	The AI Agent shall compute an OKR score (0.0–1.0) at the end of each OKR period based on achieved vs. targeted Key Results.	Defined	
REQ-WORK-006	High	Users shall be able to record weekly OKR check-ins with progress notes. AI Agent shall auto-populate check-in data from linked story completions.	Defined	
REQ-WORK-007	Medium	The system shall surface an "OKR At Risk" alert when a Key Result is tracking to fall below 0.6 score with 2 weeks remaining in the period.	Defined	

13.2 Execution Modes

ID	Priority	Requirement Description	Status	Notes
REQ-WORK-010	Critical	The system shall support Agile Sprint mode with: sprint creation, story commitment, burndown tracking, velocity history, and sprint close.	Defined	
REQ-WORK-011	Critical	The system shall support Kanban mode with: configurable columns, WIP limits, cycle time tracking, and throughput metrics.	Defined	
REQ-WORK-012	High	The system shall support Program/Milestone mode with: multi-month timelines, milestone dependencies, Gantt view, and critical path identification.	Defined	

ID	Priority	Requirement Description	Status	Notes
REQ-WORK-013	High	The system shall support Personal Routine mode with: recurring daily/weekly tasks, habit streaks, and completion rate tracking.	Defined	
REQ-WORK-014	High	Execution mode shall be configurable per WBS; a single Workspace may have WBSes in different execution modes simultaneously.	Defined	
REQ-WORK-015	Medium	The system shall provide an "Execution Mode Recommendation" from the AI Agent based on project type, team size, and timeline.	Defined	

13.3 Automation & Operations

ID	Priority	Requirement Description	Status	Notes
REQ-WORK-020	High	The system shall support user-defined Automation Rules with: event-based triggers, condition filters, and one or more actions.	Defined	
REQ-WORK-021	High	Supported automation actions shall include: create story, assign story, send notification, change status, publish to community, trigger Market search, and call webhook.	Defined	
REQ-WORK-022	High	The system shall provide pre-built automation templates for common scenarios: sprint planning, invoice follow-up, recurring tasks, contract renewal reminders.	Defined	
REQ-WORK-023	High	All automation executions shall be logged with: rule ID, trigger event, actions taken, timestamp, and success/failure status.	Defined	
REQ-WORK-024	High	The system shall support capacity planning: displaying each team member's story point commitments vs. available capacity, with configurable availability hours per week.	Defined	
REQ-WORK-025	Medium	The AI Agent shall identify and surface bottlenecks: stories blocked for > 24 hours, team members at > 120% capacity, and milestones at risk.	Defined	
REQ-WORK-026	Medium	Workflow Templates shall be creatable from any existing WBS configuration and reapplicable to new projects.	Defined	

SECTION 14 · CROWDFUNDING

Functional Requirements — Crowdfunding & Capital Pools

14.1 Campaigns

ID	Priority	Requirement Description	Status	Notes
REQ-FUND-001	Critical	Users and AOs shall be able to create Crowdfunding Campaigns with: title, description, goal amount, campaign type, duration, tiers, and media.	Defined	
REQ-FUND-002	Critical	Campaign types shall include: Donation, Reward-based, Equity, Revenue-Share, and Cooperative Membership.	Defined	
REQ-FUND-003	Critical	All campaign funds shall be held in escrow until the campaign reaches its goal (for all-or-nothing campaigns) or in rolling escrow (for keep-what-you-raise campaigns).	Defined	
REQ-FUND-004	Critical	If an all-or-nothing campaign fails to reach its goal by its end date, all backer contributions shall be fully refunded within 5 business days.	Defined	
REQ-FUND-005	High	Campaign pages shall display: progress bar, days remaining, backer count, tier availability, FAQ, and update feed.	Defined	
REQ-FUND-006	High	Campaign creators shall be able to post campaign updates viewable by all backers and publicly.	Defined	
REQ-FUND-007	High	Backers shall receive email notifications for: contribution confirmation, campaign milestone (25%/50%/75%/100% funded), campaign funded, and campaign update posted.	Defined	
REQ-FUND-008	Medium	The AI Agent shall generate an initial campaign narrative from the related Portfolio Business Case or Idea Studio concept.	Defined	
REQ-FUND-009	Medium	The system shall support recurring campaign contributions (monthly pledges) for subscription-based cooperative models.	Defined	

14.2 Capital Pools

ID	Priority	Requirement Description	Status	Notes
REQ-FUND-010	Critical	AOs and Cooperatives shall be able to create Capital Pools: shared treasuries governed by AO rules.	Defined	
REQ-FUND-011	Critical	Capital Pool transactions (deposits and withdrawals) shall require approval according to the AO's governance rules.	Defined	

ID	Priority	Requirement Description	Status	Notes
REQ-FUND-012	Critical	The Capital Pool shall maintain a full transaction ledger: every inflow, outflow, and balance change with actor, amount, reason, and governance approval reference.	Defined	
REQ-FUND-013	High	Capital Pools shall support sub-allocations: named budget buckets (e.g., "Engineering," "Marketing," "Operations") with individual limits.	Defined	
REQ-FUND-014	High	The AI Agent shall generate monthly Capital Pool reports: opening balance, all transactions, closing balance, and budget vs. actual by sub-allocation.	Defined	
REQ-FUND-015	Medium	Capital Pools shall support multiple currencies with automated conversion at the time of transaction.	Defined	

14.3 Equity & Ownership

ID	Priority	Requirement Description	Status	Notes
REQ-FUND-020	High	The system shall maintain a Cap Table for each AO with equity-issuing campaigns: stakeholder name, stake percentage, shares/units, cost basis, and acquisition date.	Defined	
REQ-FUND-021	High	Equity stakes shall be transferable between members subject to AO governance approval.	Defined	
REQ-FUND-022	High	The system shall display each member's current equity stake, dilution history, and implied value in their Portfolio.	Defined	
REQ-FUND-023	Medium	The system shall support convertible note modeling: principal, interest, conversion trigger, and discount rate, with automatic equity conversion on trigger.	Defined	

SECTION 15 · AO GOVERNANCE

Functional Requirements — Autonomous Organization Governance

15.1 AO Formation & Charter

ID	Priority	Requirement Description	Status	Notes
REQ-AO-001	Critical	Users shall be able to form an Autonomous Organization by completing a Charter: name, mission, membership rules, governance parameters, and initial member list.	Defined	
REQ-AO-002	Critical	AO Charter parameters shall include: proposal types, voting methods, quorum thresholds, supermajority thresholds, voting durations, and amendment rules.	Defined	
REQ-AO-003	Critical	Every AO shall have exactly one primary governance structure, selectable from: One-Member-One-Vote, Stake-Weighted, Hybrid (configurable mix), and Role-Based.	Defined	
REQ-AO-004	High	AO formation shall automatically provision: AO Workspace, AO Capital Pool, Member Directory, and default governance ruleset.	Defined	
REQ-AO-005	High	Charter amendments shall require a governance vote meeting the amendment threshold defined in the original charter.	Defined	
REQ-AO-006	Medium	The system shall provide AO charter templates for common organizational types: worker cooperative, investment club, open source foundation, and non-profit.	Defined	

15.2 Proposals & Voting

ID	Priority	Requirement Description	Status	Notes
REQ-AO-010	Critical	Any AO member (or member above a configurable reputation threshold) shall be able to submit a Proposal.	Defined	
REQ-AO-011	Critical	Proposal types shall include: Charter Amendment, Capital Allocation, Member Admission, Member Removal, Strategic Decision, and General Policy.	Defined	
REQ-AO-012	Critical	Each proposal shall have a configurable voting window (minimum 24 hours, maximum 30 days) during which members may vote.	Defined	

ID	Priority	Requirement Description	Status	Notes
REQ-AO-013	Critical	The system shall enforce quorum requirements: a vote shall not be valid unless a minimum percentage of eligible members have voted.	Defined	
REQ-AO-014	Critical	Voting shall be anonymous by default, with optional public voting configurable per AO.	Defined	
REQ-AO-015	High	Members shall receive in-app and email notifications when a new proposal is submitted and 48 hours before a vote closes.	Defined	
REQ-AO-016	High	The system shall display live vote tallies (if not anonymous) and a final outcome summary upon vote close.	Defined	
REQ-AO-017	High	All proposals, votes, and outcomes shall be stored in an immutable governance ledger accessible to all AO members.	Defined	
REQ-AO-018	Medium	The AI Agent shall generate a proposal summary and key considerations document for each governance proposal within 2 hours of submission.	Defined	

15.3 Membership Management

ID	Priority	Requirement Description	Status	Notes
REQ-AO-020	Critical	AO membership shall be manageable: invite, apply, approve, and remove members subject to governance rules.	Defined	
REQ-AO-021	High	AO members shall have configurable roles with different governance rights: full member (vote on all), associate (vote on select), observer (view only).	Defined	
REQ-AO-022	High	Member contribution history (stories completed, capital contributed, governance participation) shall be tracked and displayed in the AO Member Directory.	Defined	
REQ-AO-023	Medium	The system shall support member reputation scores within an AO based on: contribution volume, quality (peer ratings), governance participation, and tenure.	Defined	

SECTION 16 · AI AGENT

Functional Requirements — AI Agent & Execution

16.1 Core AI Agent Capabilities

ID	Priority	Requirement Description	Status	Notes
REQ-AI-001	Critical	The platform shall provide an AI Agent accessible from every module, responsive to member queries and platform events.	Defined	
REQ-AI-002	Critical	The AI Agent shall be invokable via: direct chat, @kogi mention in any room, and automated platform triggers.	Defined	
REQ-AI-003	Critical	The AI Agent shall have read access to the invoking user's Portfolio, WBS, and Workspace data to provide contextually relevant responses.	Defined	
REQ-AI-004	Critical	AI Agent responses shall be clearly attributed as AI-generated and shall not be presented as authoritative decisions without member confirmation.	Defined	
REQ-AI-005	High	The AI Agent shall maintain conversation context within a session and allow members to reference previous messages in the same thread.	Defined	
REQ-AI-006	High	The AI Agent shall cite the data sources (portfolio items, market listings, external sources) used in its responses.	Defined	
REQ-AI-007	High	Members shall be able to give the AI Agent explicit permissions to take actions on their behalf (e.g., create story, send message, trigger search).	Defined	
REQ-AI-008	Medium	The AI Agent shall provide a reasoning summary ("I did this because...") when taking autonomous actions on a member's behalf.	Defined	

16.2 Autonomous Actions

ID	Priority	Requirement Description	Status	Notes
REQ-AI-010	High	The AI Agent shall autonomously create and assign WBS stories based on incoming events when an automation rule is configured to do so.	Defined	
REQ-AI-011	High	The AI Agent shall generate draft reports, summaries, and documents from Portfolio and WBS data without requiring the member to specify individual data points.	Defined	

ID	Priority	Requirement Description	Status	Notes
REQ-AI-012	High	The AI Agent shall proactively alert members to: approaching deadlines, blocked stories (>24h), budget overruns, OKRs at risk, and contract renewals.	Defined	
REQ-AI-013	High	The AI Agent shall match unassigned WBS stories to Market providers based on story requirements and provider capabilities.	Defined	
REQ-AI-014	Medium	The AI Agent shall propose sprint plans based on backlog priority, historical velocity, and team capacity.	Defined	
REQ-AI-015	Medium	The AI Agent shall generate AI-synthesized retrospective documents from sprint completion data.	Defined	
REQ-AI-016	Medium	The AI Agent shall recommend new Community connections and Spaces based on portfolio activity and skill profile.	Defined	

16.3 AI Executor Types

ID	Priority	Requirement Description	Status	Notes
REQ-AI-020	High	The system shall support five Executor types for WBS story assignment: AI Agent (fully automated), Human (manual), Hybrid (AI-assisted human), Machine (scripted system), and Custom (third-party integration).	Defined	
REQ-AI-021	High	AI Agent Executors shall update story status, log progress notes, and flag blockers without human intervention when configured to do so.	Defined	
REQ-AI-022	Medium	The system shall track AI Agent execution quality: story completion rate, time-to-complete, member override rate, and satisfaction rating.	Defined	

SECTION 17 · NON-FUNCTIONAL

Non-Functional Requirements

17.1 Performance

ID	Priority	Requirement Description	Status	Notes
REQ-NFR-001	Critical	API response time for read operations shall be < 200ms at p95 under normal load (< 10,000 concurrent users).	Defined	
REQ-NFR-002	Critical	API response time for write operations shall be < 500ms at p95 under normal load.	Defined	
REQ-NFR-003	Critical	Real-time message delivery (chat) shall achieve < 500ms end-to-end latency at p95.	Defined	
REQ-NFR-004	Critical	The platform shall support 100,000 concurrent users in v1.0 with horizontal scaling to 1M+ concurrent users.	Defined	
REQ-NFR-005	High	Full-text search results shall return within 1 second for queries against up to 10M listings.	Defined	
REQ-NFR-006	High	AI Agent text responses shall begin streaming within 3 seconds of query submission.	Defined	
REQ-NFR-007	High	Page load time for web client (SPA route transitions) shall be < 1 second at p95 on a 25Mbps connection.	Defined	
REQ-NFR-008	Medium	File uploads up to 50MB shall complete within 30 seconds on a 10Mbps connection.	Defined	
REQ-NFR-009	Medium	Export operations (portfolio export, transaction history) for up to 10,000 records shall complete within 60 seconds.	Defined	

17.2 Availability & Reliability

ID	Priority	Requirement Description	Status	Notes
REQ-NFR-010	Critical	The platform shall achieve 99.9% uptime (SLA) for all Critical-tier services, measured monthly.	Defined	
REQ-NFR-011	Critical	The platform shall not have planned maintenance downtime exceeding 2 hours per month, announced at least 72 hours in advance.	Defined	
REQ-NFR-012	Critical	The system shall implement automated failover for all stateful services with < 30 seconds Recovery Time Objective (RTO).	Defined	

ID	Priority	Requirement Description	Status	Notes
REQ-NFR-013	Critical	Recovery Point Objective (RPO) for financial transaction data shall not exceed 1 minute.	Defined	
REQ-NFR-014	High	All database data shall be replicated across at minimum 3 geographic availability zones.	Defined	
REQ-NFR-015	High	The system shall perform automated backups every 6 hours with a 30-day retention period.	Defined	
REQ-NFR-016	High	The platform shall implement graceful degradation: if a non-critical module fails, the rest of the platform shall continue to function.	Defined	

17.3 Scalability

ID	Priority	Requirement Description	Status	Notes
REQ-NFR-020	Critical	All platform services shall be horizontally scalable: new instances shall be addable without downtime.	Defined	
REQ-NFR-021	High	The platform shall support auto-scaling based on: CPU utilization (> 70%), memory utilization (> 80%), and request queue depth (> 1,000 pending).	Defined	
REQ-NFR-022	High	The Event Bus shall handle peak throughput of 100,000 events/second without message loss.	Defined	
REQ-NFR-023	Medium	The system shall support database sharding for Portfolio and WBS data beyond 100M records.	Defined	

17.4 Usability & Accessibility

ID	Priority	Requirement Description	Status	Notes
REQ-NFR-030	Critical	All web and mobile interfaces shall conform to WCAG 2.1 Level AA accessibility standards.	Defined	
REQ-NFR-031	Critical	The platform shall support keyboard-only navigation for all core workflows in the web client.	Defined	
REQ-NFR-032	High	The web client shall support screen readers (NVDA, JAWS, VoiceOver) for all critical user flows.	Defined	
REQ-NFR-033	High	The platform shall provide platform UI in English as the primary language, with internationalization (i18n) framework in place for future language additions.	Defined	
REQ-NFR-034	High	The web client shall be responsive, providing a usable experience at viewport widths from 320px to 4K.	Defined	

ID	Priority	Requirement Description	Status	Notes
REQ-NFR-035	Medium	User onboarding shall guide new members through first-time setup in < 5 minutes with progressive disclosure of advanced features.	Defined	
REQ-NFR-036	Medium	All user-facing error messages shall be in plain language with a suggested resolution action.	Defined	

SECTION 18 · SECURITY

Security & Privacy Requirements

18.1 Authentication & Authorization

ID	Priority	Requirement Description	Status	Notes
REQ-SEC-001	Critical	All API endpoints shall require authentication unless explicitly designated public (e.g., public listing pages).	Defined	
REQ-SEC-002	Critical	Authentication tokens (JWT) shall have a maximum lifetime of 24 hours and shall be refreshable for active sessions.	Defined	
REQ-SEC-003	Critical	The system shall invalidate all active sessions upon password change or account compromise report.	Defined	
REQ-SEC-004	Critical	All authorization checks shall be enforced server-side; client-side permission checks shall be treated as UX hints only.	Defined	
REQ-SEC-005	Critical	The system shall implement rate limiting on authentication endpoints: maximum 10 failed login attempts per IP per 15 minutes before temporary lockout.	Defined	
REQ-SEC-006	High	The system shall log all authentication events (login, logout, token refresh, MFA) with IP, user agent, and timestamp.	Defined	
REQ-SEC-007	High	API keys for Exchange and AI integration shall be stored encrypted at rest and never transmitted in plaintext.	Defined	

18.2 Data Security

ID	Priority	Requirement Description	Status	Notes
REQ-SEC-010	Critical	All data at rest shall be encrypted using AES-256 or equivalent.	Defined	
REQ-SEC-011	Critical	All data in transit shall be encrypted using TLS 1.2 or higher; TLS 1.0 and 1.1 shall be disabled.	Defined	
REQ-SEC-012	Critical	Financial transaction data and personally identifiable information (PII) shall be encrypted with a separate key management system (KMS).	Defined	
REQ-SEC-013	Critical	The platform shall undergo an annual third-party penetration test with findings remediated within 30 days (Critical) or 90 days (High).	Defined	

ID	Priority	Requirement Description	Status	Notes
REQ-SEC-014	High	The system shall implement a Web Application Firewall (WAF) protecting against OWASP Top 10 vulnerability categories.	Defined	
REQ-SEC-015	High	SQL injection, XSS, CSRF, and injection attacks shall be prevented through parameterized queries, input validation, and output encoding.	Defined	
REQ-SEC-016	High	Secrets, API keys, and credentials shall never be stored in source code repositories; all secrets shall be managed via a secrets management service.	Defined	

18.3 Privacy

ID	Priority	Requirement Description	Status	Notes
REQ-SEC-020	Critical	The platform shall have a published Privacy Policy clearly stating: what data is collected, how it is used, who it is shared with, and how users can exercise their rights.	Defined	
REQ-SEC-021	Critical	Users shall be able to access all personal data the platform holds about them within 30 days of request.	Defined	
REQ-SEC-022	Critical	Users shall be able to request deletion of their personal data. Deletion shall be complete within 30 days except where retention is required by law.	Defined	
REQ-SEC-023	Critical	The AI Agent shall not use one user's private Portfolio or workspace data to inform responses to another user without explicit cross-sharing consent.	Defined	
REQ-SEC-024	High	The platform shall implement data minimization: only the data necessary for each feature shall be collected and stored.	Defined	
REQ-SEC-025	High	Third-party integrations (payment processors, AI providers) shall be governed by Data Processing Agreements (DPAs).	Defined	
REQ-SEC-026	Medium	Users shall be able to opt out of non-essential data collection (analytics, behavioral tracking) while retaining full platform access.	Defined	

SECTION 19 · COMPLIANCE

Compliance & Regulatory Requirements

19.1 Financial Compliance

ID	Priority	Requirement Description	Status	Notes
REQ-COMP-001	Critical	All payment processing shall comply with PCI DSS Level 1 standards; card data shall never touch Kogi servers directly.	Defined	
REQ-COMP-002	Critical	The Exchange System shall comply with applicable money transmission laws in each operating jurisdiction.	Defined	
REQ-COMP-003	Critical	Equity-based crowdfunding features shall comply with applicable securities laws (e.g., SEC Regulation Crowdfunding in the US, similar regimes globally).	Defined	
REQ-COMP-004	High	The platform shall implement KYC (Know Your Customer) verification for users transacting above configurable thresholds ($1,000 in any 30-day period by default).	Defined	
REQ-COMP-005	High	The platform shall implement AML (Anti-Money Laundering) transaction monitoring and flag suspicious patterns for compliance review.	Defined	
REQ-COMP-006	High	The platform shall generate IRS Form 1099-K, 1099-NEC, and 1099-DIV data for applicable US-based transactions.	Defined	
REQ-COMP-007	Medium	The platform shall implement OFAC sanctions list screening for all financial transactions.	Defined	

19.2 Data Protection Compliance

ID	Priority	Requirement Description	Status	Notes
REQ-COMP-010	Critical	The platform shall comply with GDPR for all users in the European Economic Area, including: lawful basis for processing, consent management, data subject rights, and breach notification.	Defined	
REQ-COMP-011	Critical	The platform shall comply with CCPA for California residents, including: right to know, right to delete, and right to opt out of sale.	Defined	
REQ-COMP-012	High	The platform shall maintain Records of Processing Activities (ROPA) as required by GDPR Article 30.	Defined	

ID	Priority	Requirement Description	Status	Notes
REQ-COMP-013	High	Data breach notification to affected users and relevant supervisory authorities shall occur within 72 hours of confirmed breach discovery.	Defined	
REQ-COMP-014	Medium	The platform shall implement Privacy by Design and Privacy by Default principles in all new feature development.	Defined	

19.3 Labor & Cooperative Compliance

ID	Priority	Requirement Description	Status	Notes
REQ-COMP-020	Medium	The AO Governance module shall provide charter templates compliant with cooperative corporation laws in the US (state-by-state) and the EU.	Defined	
REQ-COMP-021	Medium	The system shall not misclassify gig workers and shall provide functionality for worker-owned cooperative structures as an alternative to traditional employer-platform models.	Defined	
REQ-COMP-022	Low	The platform shall provide tooling to assist cooperatives in maintaining records required by applicable cooperative statutes (membership rolls, minutes, financial statements).	Defined	

SECTION 20 · INTEGRATION

Integration & API Requirements

20.1 REST API

ID	Priority	Requirement Description	Status	Notes
REQ-INT-001	Critical	The platform shall expose a fully documented public REST API for all core modules accessible to third-party developers.	Defined	
REQ-INT-002	Critical	All API endpoints shall follow RESTful conventions: resource-based URLs, appropriate HTTP methods, standard status codes, and consistent error response format.	Defined	
REQ-INT-003	Critical	API versioning shall be managed via URL path prefix (e.g., /api/v1/, /api/v2/) with a minimum 12-month deprecation notice before removing versions.	Defined	
REQ-INT-004	High	The platform shall provide OpenAPI 3.0 specification documents for all API endpoints, auto-generated from code.	Defined	
REQ-INT-005	High	API responses shall include pagination (cursor-based for large datasets), filtering, sorting, and field selection capabilities.	Defined	
REQ-INT-006	High	API rate limits shall be enforced per authenticated user: 1,000 requests/minute standard, 10,000/minute for verified enterprise integrations.	Defined	

20.2 Webhooks & Events

ID	Priority	Requirement Description	Status	Notes
REQ-INT-010	High	The platform shall support outbound webhooks: third-party systems can subscribe to platform events and receive HTTP POST notifications.	Defined	
REQ-INT-011	High	Webhook deliveries shall include a cryptographic signature header for request authentication by the receiving system.	Defined	
REQ-INT-012	High	Failed webhook deliveries shall be retried with exponential backoff for up to 24 hours.	Defined	
REQ-INT-013	Medium	The platform shall provide a webhook event log showing delivery status, response codes, and retry history.	Defined	

20.3 External Integrations

ID	Priority	Requirement Description	Status	Notes
REQ-INT-020	High	The platform shall support OAuth 2.0-based connections to external services: Google Calendar, Google Drive, Slack, GitHub, Notion, and Stripe.	Defined	
REQ-INT-021	High	GitHub integration shall enable: linking commits and PRs to WBS stories, auto-updating story status on PR merge, and displaying code contribution stats in the AO contributor profile.	Defined	
REQ-INT-022	High	Calendar integrations (Google Calendar, Apple Calendar) shall enable two-way sync of scheduled tasks, milestones, and Workspace events.	Defined	
REQ-INT-023	Medium	The platform shall support Zapier and Make (formerly Integromat) as automation middleware, exposing platform triggers and actions.	Defined	
REQ-INT-024	Medium	The platform shall support importing portfolio data from: Notion, Asana, Jira, Trello, Basecamp, and Linear via structured CSV or direct API.	Defined	

20.4 gRPC & Internal

ID	Priority	Requirement Description	Status	Notes
REQ-INT-030	High	Inter-service communication between platform microservices shall use gRPC with Protocol Buffer schemas for performance-critical paths.	Defined	
REQ-INT-031	High	All gRPC service definitions shall be maintained in the /proto directory of the monorepo and versioned with the same policy as REST APIs.	Defined	
REQ-INT-032	Medium	The Zig kernel shall expose a C-compatible FFI interface for integration with the Go API gateway.	Defined	

SECTION 21 · DATA

Data Requirements

21.1 Data Integrity

ID	Priority	Requirement Description	Status	Notes
REQ-DATA-001	Critical	All financial transaction records shall be immutable: once written, they shall not be editable or deletable, only corrected via compensating entries.	Defined	
REQ-DATA-002	Critical	The system shall use optimistic concurrency control for concurrent edits to shared resources (Portfolio items, stories) and surface merge conflicts to users.	Defined	
REQ-DATA-003	Critical	All database operations affecting multiple tables shall be wrapped in ACID transactions.	Defined	
REQ-DATA-004	High	The system shall maintain full audit logs for: governance decisions, permission changes, financial transactions, and AI Agent autonomous actions.	Defined	
REQ-DATA-005	High	Deleted items (stories, portfolio items) shall be soft-deleted with a 30-day recovery window before permanent removal.	Defined	

21.2 Data Retention

ID	Priority	Requirement Description	Status	Notes
REQ-DATA-010	Critical	Financial transaction records shall be retained for a minimum of 7 years to meet standard accounting and tax requirements.	Defined	
REQ-DATA-011	Critical	Governance votes and AO decisions shall be retained for the lifetime of the AO plus 7 years.	Defined	
REQ-DATA-012	High	User-generated content (messages, stories, documents) shall be retained for the lifetime of the account plus a 30-day post-deletion window.	Defined	
REQ-DATA-013	High	Authentication logs shall be retained for 12 months.	Defined	
REQ-DATA-014	Medium	AI Agent interaction logs shall be retained for 90 days and used only for quality improvement and debugging with user consent.	Defined	

21.3 Portability & Interoperability

ID	Priority	Requirement Description	Status	Notes
REQ-DATA-020	High	Users shall be able to export their complete Portfolio data in JSON format including all nested items, resources, and metadata.	Defined	
REQ-DATA-021	High	WBS data shall be exportable in: JSON, CSV (flat), and Markdown formats.	Defined	
REQ-DATA-022	High	Financial transaction history shall be exportable in: CSV, PDF (formatted statement), and QFX/OFX (for import into accounting software).	Defined	
REQ-DATA-023	Medium	Community message history for Spaces owned by the user shall be exportable in JSON format.	Defined	
REQ-DATA-024	Medium	The platform shall support bulk data import for onboarding: portfolio structure from JSON, stories from CSV, and team members via CSV.	Defined	

SECTION 22 · GLOSSARY

Glossary

Term	Definition
AO	Autonomous Organization — a Kogi entity with its own charter, governance, capital pool, and member base.
Artifact	A document, file, image, or data object stored within a Portfolio container.
Backlog	The collection of stories that have been defined but not yet committed to a sprint or work period.
Capital Pool	A shared treasury managed by an AO or Cooperative, governed by that entity's rules.
Container	A structural item attached to a Portfolio item to organize sub-documents: Binder, Book, Notepad, Folder, or Briefcase.
Cooperative	A member-owned organization with democratic governance; a specific type of AO with cooperative legal structure.
Crowdfunding Campaign	A time-limited fundraising campaign attached to a Portfolio item, supporting multiple funding models.
Escrow	Funds held by the Exchange System on behalf of two parties pending delivery confirmation.
Executor	An entity assigned to execute a WBS story: AI Agent, Human, Hybrid, Machine, or Custom.
Gig	A discrete, on-demand task or delivery job listed in the Market and executed by a Gig Worker.
Idea Studio	The Kogi module for capturing and developing ideas through a five-stage innovation pipeline.
Kernel	The central platform orchestrator managing module registration, event bus, and provisioning.
Key Result	A measurable indicator of progress toward an Objective in the OKR framework.
Listing	A Market entry representing a service, gig, product, talent profile, tool, or resource offered for sale or hire.
OKR	Objectives and Key Results — a goal-setting framework used in the Work & Strategy module.
Portfolio	The primary organizational structure for a user or AO's work, assets, and investments.
Program	A top-level Portfolio item representing a long-running initiative containing multiple Projects.
Proposal	A governance item in an AO that members vote on according to the charter's rules.

Term	Definition
Provisioning	The automated process of creating a new user's Account, Workspace, and Portfolio upon registration.
Room	A communication context within a Community Space: General, Live, Voice, Video, Showcase, Event, or Q&A.
Service Credits	Platform-native non-monetary value units earned through contribution and spendable in the Market.
Space	A Community hub organized around a topic, industry, project, or organizational membership.
Sprint	A time-boxed iteration (1–4 weeks) in Agile execution mode, containing a committed set of stories.
Story	The primary unit of work in the Kogi WBS system. Comes in 30+ types. Has a status, priority, and executor.
Wallet	An Exchange System account per user holding fiat balance, service credits, and equity holdings.
WBS	Work Breakdown Structure — a hierarchical decomposition of work into: Work Package → Theme → Initiative → Epic → Story → Task.
Workspace	A user's or organization's primary environment on the Kogi platform, providing access to all modules.

Kogi Platform Requirements Specification

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